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EXP 6: ENUMERATE WEB SERVER INFORMATION

AIM:

To gather detailed configuration and version information about a target web server using enumeration tools such as Nmap, curl, and Gobuster.

TOOLS REQUIRED:

1. Kali Linux – Operating system containing the enumeration tools.
2. Metasploitable2 – Vulnerable target machine hosting the web server.
3. Nmap – To identify open ports and enumerate supported HTTP methods.
4. curl – To inspect HTTP response headers and request common web files.
5. Gobuster – To discover common directories and files using a wordlist.
6. Web browser – To check the /uploads/ path.
7. OpenSSL – For TLS inspection if HTTPS is available (not used because port 443 was closed).

PROCEDURE:

1. Start Kali Linux and Metasploitable2 and verify network connectivity using ping.

   ping -c 4 192.168.56.101

2. Inspect the HTTP response headers using curl to identify the server and PHP version.

   curl -X OPTIONS -i http://192.168.56.101

3. Use Nmap's HTTP methods script to identify methods supported by the web server.

   nmap -p80 --script=http-methods 192.168.56.101

4. Check whether the /uploads/ path is present by opening it in a web browser.

   http://192.168.56.101/uploads/

5. Request robots.txt and sitemap.xml to check whether these common web files are available.

   curl -i http://192.168.56.101/robots.txt
   curl -i http://192.168.56.101/sitemap.xml

6. Use Gobuster to enumerate common web paths with the DIRB common wordlist.

   gobuster dir -u http://192.168.56.101 -w /usr/share/wordlists/dirb/common.txt

7. Check whether HTTPS is listening on port 443. If it is closed, TLS cipher enumeration cannot be performed.

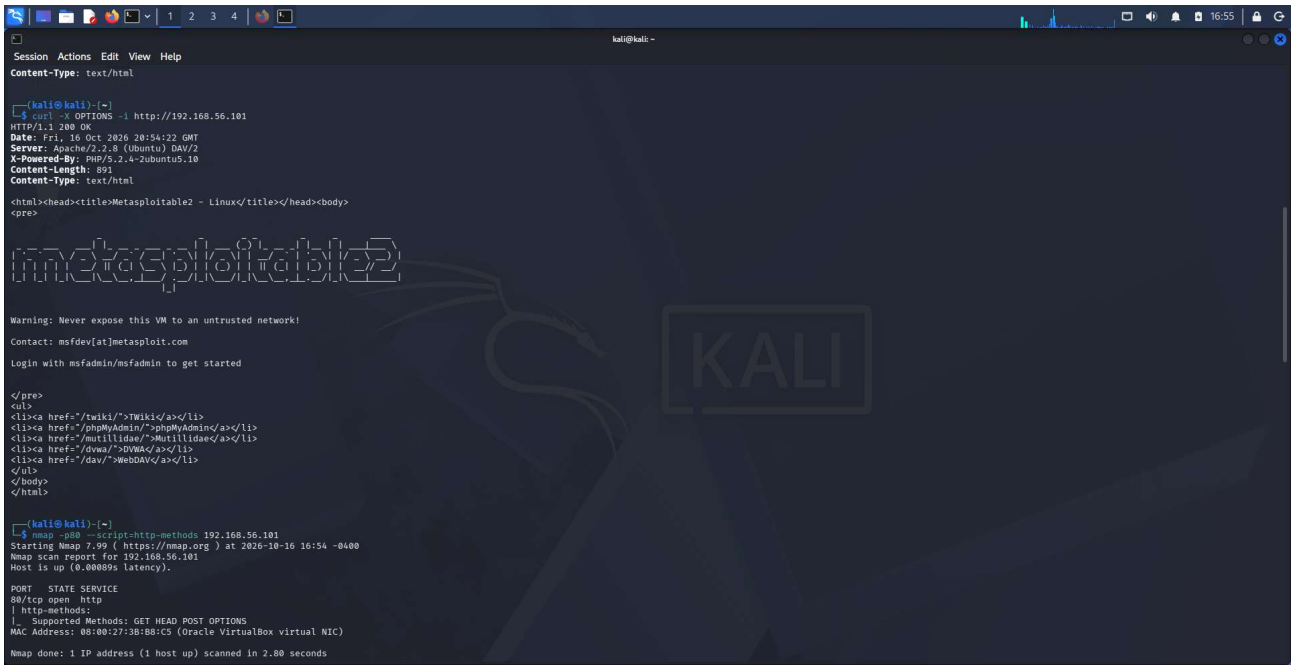
   nmap -p443 192.168.56.101

8. Record the outputs obtained, analyse the web server information, and document potential security observations.

OBSERVATION:

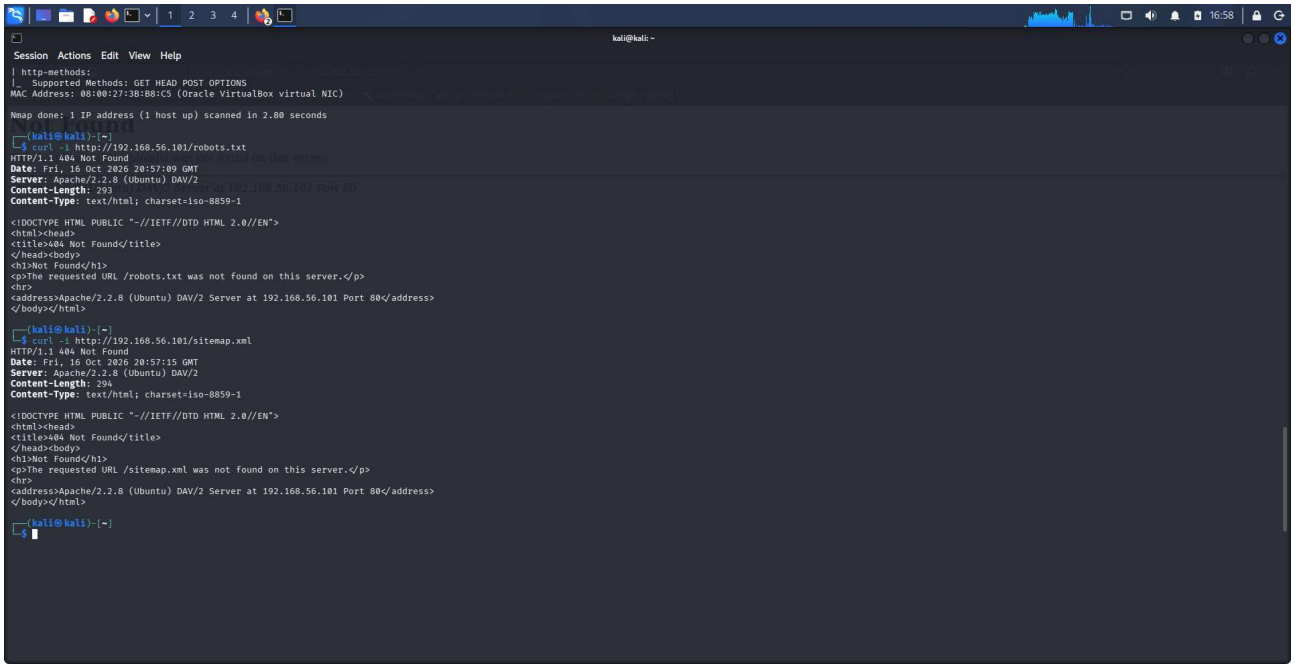
TEST PERFORMED	OBSERVATION
HTTP response header inspection using curl	Server returned HTTP/1.1 200 OK. Headers identified Apache/2.2.8 (Ubuntu) DAV/2 and PHP/5.2.4-2ubuntu5.10. Content type was text/html.
Nmap HTTP method enumeration	Port 80/tcp was open. Supported methods reported by Nmap: GET, HEAD, POST, and OPTIONS.
Check /uploads/ path	The browser displayed Not Found. The requested /uploads/ path returned 404; directory indexing was not demonstrated.
Check robots.txt and sitemap.xml	Both /robots.txt and /sitemap.xml returned HTTP/1.1 404 Not Found.
Directory and file enumeration using Gobuster	Discovered /.htpasswd, /.htaccess, /.hta, /cgi-bin/, and /server-status with status 403; /dav, /phpMyAdmin, /test, and /twiki with status 301; and /index, /index.php, /phpinfo, and /phpinfo.php with status 200.
HTTPS port check using Nmap	Port 443/tcp was closed while the target host was up. SSL/TLS cipher enumeration was therefore not performed.

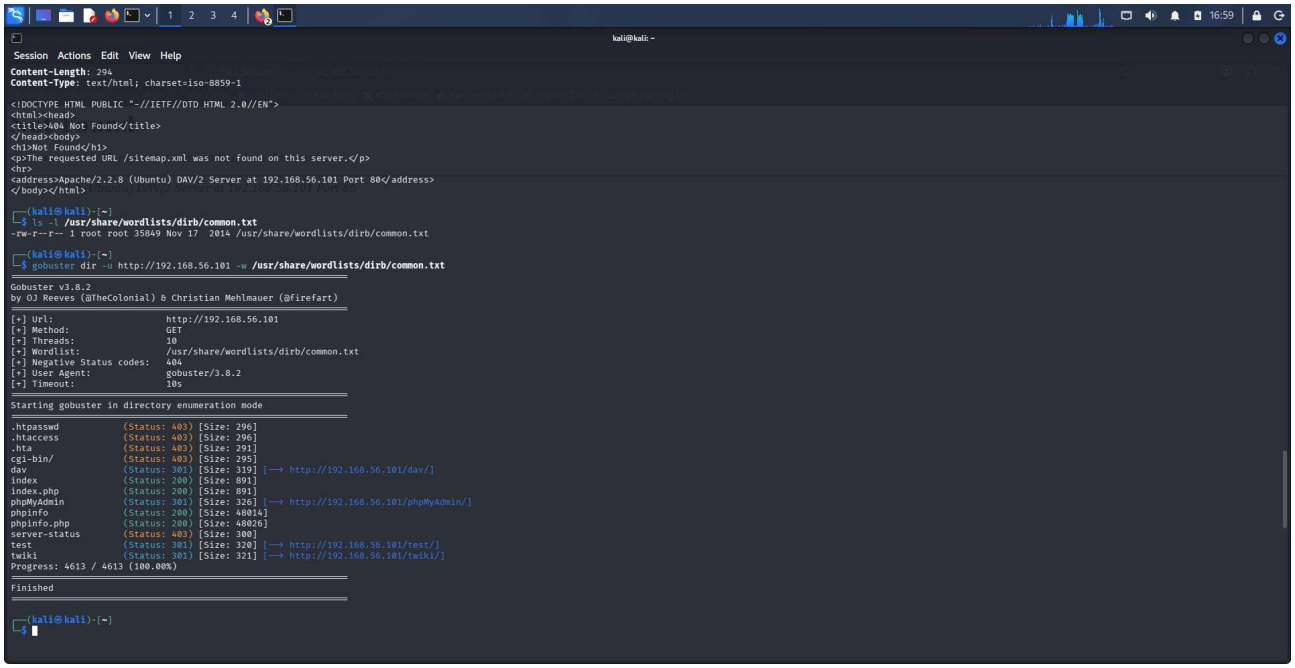
OUTPUT:

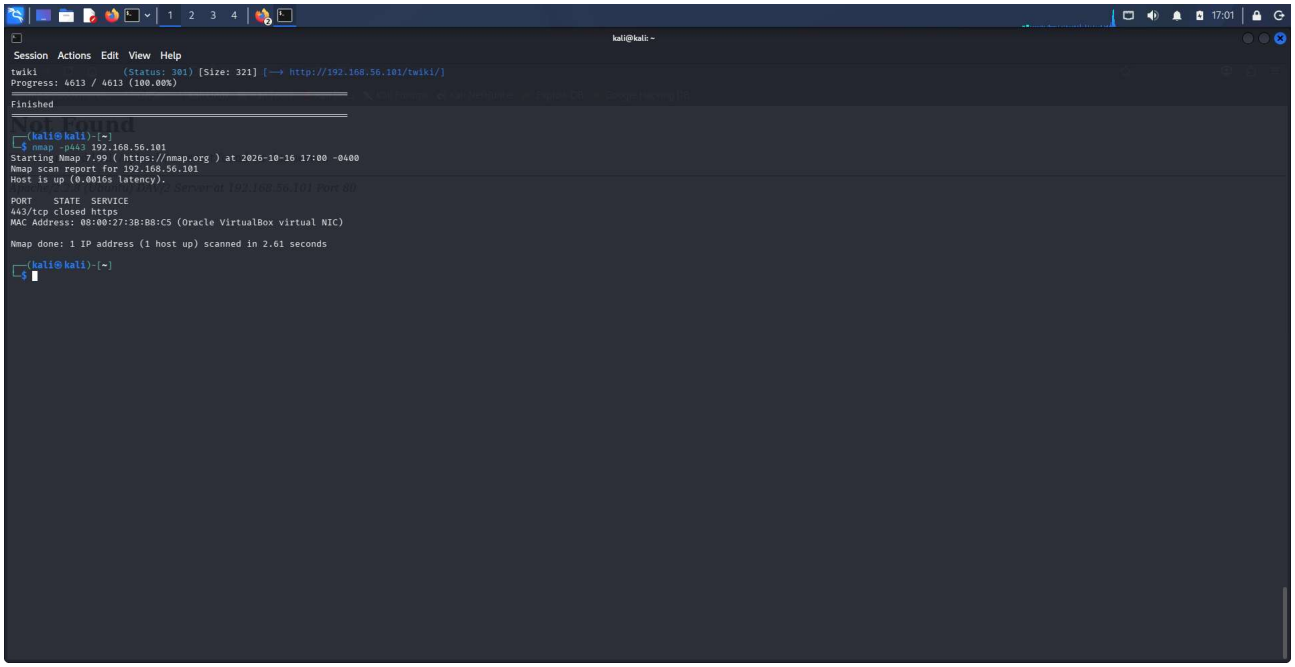

kali@kali: ~
Session Actions Edit View Help
Content-Type: text/html
┌──(kali㉿kali)-[~]
└─$ curl -X OPTIONS -i http://192.168.56.101
HTTP/1.1 200 OK
Date: Fri, 16 Oct 2026 20:54:22 GMT
Server: Apache/2.2.8 (Ubuntu) DAV/2
X-Powered-By: PHP/5.2.4-2ubuntu5.10
Content-Length: 891
Content-Type: text/html
<html><head><title>Metasploitable2 - Linux</title></head><body>
<pre>
Warning: Never expose this VM to an untrusted network!
Contact: msfdev[at]metasploit.com
Login with msfadmin/msfadmin to get started
</pre>

TWiki
phpMyAdmin
Mutillidae
DVWA
WebDAV

</body>
</html>
┌──(kali㉿kali)-[~]
└─$ nmap -p80 --script=http-methods 192.168.56.101
Starting Nmap 7.99 (https://nmap.org) at 2026-10-16 16:54 -0400
Nmap scan report for 192.168.56.101
Host is up (0.00089s latency).
PORT STATE SERVICE
80/tcp open http
| http-methods:
|_ Supported Methods: GET HEAD POST OPTIONS
MAC Address: 08:00:27:3B:B8:C5 (Oracle VirtualBox virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 2.80 seconds


kali@kali: ~
Session Actions Edit View Help
| http-methods:
|_ Supported Methods: GET HEAD POST OPTIONS
MAC Address: 08:00:27:3B:B8:C5 (Oracle VirtualBox virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 2.80 seconds
┌──(kali㉿kali)-[~]
└─$ curl -i http://192.168.56.101/robots.txt
HTTP/1.1 404 Not Found
Date: Fri, 16 Oct 2026 20:57:09 GMT
Server: Apache/2.2.8 (Ubuntu) DAV/2
Content-Length: 293
Content-Type: text/html; charset=iso-8859-1
<!DOCTYPE HTML PUBLIC "-//IETF//DTD HTML 2.0//EN">
<html><head>
<title>404 Not Found</title>
</head><body>
<h1>Not Found</h1>
The requested URL /robots.txt was not found on this server.
<hr>
<address>Apache/2.2.8 (Ubuntu) DAV/2 Server at 192.168.56.101 Port 80</address>
</body></html>
┌──(kali㉿kali)-[~]
└─$ curl -i http://192.168.56.101/sitemap.xml
HTTP/1.1 404 Not Found
Date: Fri, 16 Oct 2026 20:57:15 GMT
Server: Apache/2.2.8 (Ubuntu) DAV/2
Content-Length: 294
Content-Type: text/html; charset=iso-8859-1
<!DOCTYPE HTML PUBLIC "-//IETF//DTD HTML 2.0//EN">
<html><head>
<title>404 Not Found</title>
</head><body>
<h1>Not Found</h1>
The requested URL /sitemap.xml was not found on this server.
<hr>
<address>Apache/2.2.8 (Ubuntu) DAV/2 Server at 192.168.56.101 Port 80</address>
</body></html>
┌──(kali㉿kali)-[~]
└─$





RESULT:

Web server enumeration was carried out using Nmap, curl, and Gobuster in an authorised lab environment to examine HTTP response headers, supported methods, common web paths, and HTTPS availability. The target reported Apache/2.2.8 (Ubuntu) DAV/2 and PHP/5.2.4-2ubuntu5.10. Gobuster identified paths returning 200, 301, and 403 status codes, while /uploads/, robots.txt, and sitemap.xml returned 404 Not Found. Port 443 was closed.